MyEconLab Snapshot

SHORT RUN AND LONG RUN

The main goal of this chapter is to explore the influences on a firm's costs. The key influence on cost is the quantity of output that the firm produces per period. The greater the output rate, the higher is the total cost of production. But the effect of a change in production on cost depends on how soon the firm wants to act. A firm that plans to change its output rate tomorrow has fewer options than a firm that plans ahead and intends to change its production six months from now.

To study the relationship between a firm's output decision and its costs, we distinguish between two decision time frames:

- The short run
- The long run

The Short Run: Fixed Plant

Short run
The time frame in which the quantities of some resources are fixed. In the short run, a firm can usually change the quantity of labor it uses but not its technology and quantity of capital.

The **short run** is the time frame in which the quantities of some resources are fixed. For most firms, the fixed resources are the firm's technology and capital—its equipment and buildings. The management organization is also fixed in the short run. The fixed resources that a firm uses are its *fixed factors of production* and the resources that it can vary are its *variable factors of production*. The collection of fixed resources is the firm's *plant*. So in the short run, a firm's plant is fixed.

Sam's Smoothies' plant is its blenders, refrigerators, and shop. Sam's cannot change these inputs in the short run. An electric power utility can't change the number of generators it uses in the short run. An airport can't change the number of runways, terminal buildings, and traffic-control facilities in the short run.

To increase output in the short run, a firm must increase the quantity of variable factors it uses. Labor is usually the variable factor of production. To produce more smoothies, Sam must hire more labor. Similarly, to increase the production of electricity, a utility must hire more engineers and run its generators for longer hours. To increase the volume of traffic it handles, an airport must hire more check-in clerks, cargo handlers, and air-traffic controllers.

Short-run decisions are easily reversed. A firm can increase or decrease output in the short run by increasing or decreasing the number of labor hours it hires.

The Long Run: Variable Plant

Long run
The time frame in which the quantities of *all* resources can be varied.

The **long run** is the time frame in which the quantities of *all* resources can be varied. That is, the long run is a period in which the firm can change its *plant*.

To increase output in the long run, a firm can increase the size of its plant. Sam's Smoothies can install more blenders and refrigerators and increase the size of its shop. An electric power utility can install more generators. And an airport can build more runways, terminals, and traffic-control facilities.

Long-run decisions are not easily reversed. Once a firm buys a new plant, its resale value is usually much less than the amount the firm paid for it. The fall in value is economic depreciation. It is called a *sunk cost* to emphasize that it is irrelevant to the firm's decisions. Only the short-run cost of changing its labor inputs and the long-run cost of changing its plant size are relevant to a firm's decisions.

We're going to study costs in the short run and the long run. We begin with the short run and describe the limits to the firm's production possibilities.

14.2 SHORT-RUN PRODUCTION

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To increase the output of a fixed plant, a firm must increase the quantity of labor it employs. We describe the relationship between output and the quantity of labor employed by using three related concepts:

- Total product
- Marginal product
- Average product

Total Product

Total product (*TP*) is the total quantity of a good produced in a given period. Total product is an output *rate*—the number of units produced per unit of time (for example, per hour, day, or week). Total product changes as the quantity of labor employed increases and we illustrate this relationship as a total product schedule and total product curve like those in Figure 14.2. The total product schedule (the table below the graph) lists the maximum quantities of smoothies per hour that Sam can produce with her existing plant at each quantity of labor. Points *A* through *H* on the *TP* curve correspond to the columns in the table.

Total product
The total quantity of a good produced in a given period.

FIGURE 14.2

Total Product Schedule and Total Product Curve

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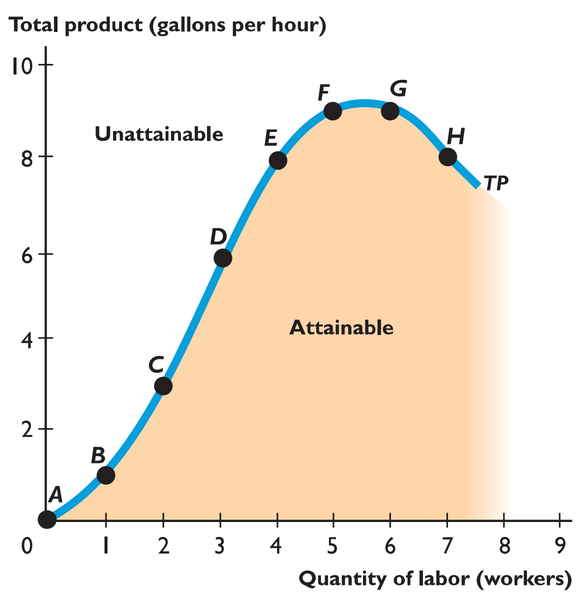


Quantity of labor (workers)	0	1	2	3	4	5	6	7
Total product (gallons per hour)	0	1	3	6	8	9	9	8
	A	*B*	*C*	*D*	*E*	*F*	*G*	*H*

The total product schedule shows how the quantity of smoothies that Sam's can produce changes as the quantity of labor employed changes. In column *C*, Sam's employs 2 workers and can produce 3 gallons of smoothies an hour.

The total product curve, *TP*, graphs the data in the table. Points *A* through *H* on the curve correspond to the columns of the table. The total product curve separates attainable outputs from unattainable outputs. Points below the *TP* curve are inefficient. Points on the *TP* curve are efficient.

Like the *production possibilities frontier* (see Chapter 3, p. 60), the total product curve separates attainable outputs from unattainable outputs. All the points that lie above the curve are unattainable. Points that lie below the curve, in the orange area, are attainable, but they are inefficient: They use more labor than is necessary to produce a given output. Only the points *on* the total product curve are efficient.

Marginal Product

Marginal product
The change in total product that results from a one-unit increase in the quantity of labor employed.

Marginal product (*MP*) is the change in total product that results from a one-unit increase in the quantity of labor employed. It tells us the contribution to total product of adding one additional worker. When the quantity of labor increases by more than one worker, we calculate marginal product as

Marginal product = Change in total product ÷ Change in quanity of labor.

Figure 14.3 shows Sam's Smoothies' marginal product curve, *MP*, and its relationship with the total product curve. You can see that as the quantity of labor increases from 1 to 3 workers, marginal product increases. But as more than 3 workers are employed, marginal product decreases. When the seventh worker is employed, marginal product is negative.

Notice that the steeper the slope of the total product curve in part (a), the greater is marginal product in part (b). And when the total product curve turns downward in part (a), marginal product is negative in part (b).

The total product curve and marginal product curve in Figure 14.3 incorporate a feature that is shared by all production processes in firms as different as the Ford Motor Company, Jim's Barber Shop, and Sam's Smoothies:

- Increasing marginal returns initially
- Decreasing marginal returns eventually

Increasing Marginal Returns

Increasing marginal returns
When the marginal product of an additional worker exceeds the marginal product of the previous worker.

Increasing marginal returns occur when the marginal product of an additional worker exceeds the marginal product of the previous worker. The source of increasing marginal returns is increased specialization and greater division of labor in the production process.

For example, if Samantha employs just one worker, that person must learn all the aspects of making smoothies: running the blender, cleaning it, fixing breakdowns, buying and checking the fruit, and serving the customers. That one person must perform all these tasks.

If Samantha hires a second person, the two workers can specialize in different parts of the production process. As a result, two workers can produce more than twice as much as one worker. The marginal product of the second worker is greater than the marginal product of the first worker. Marginal returns are increasing. Most production processes experience increasing marginal returns initially.

Decreasing Marginal Returns

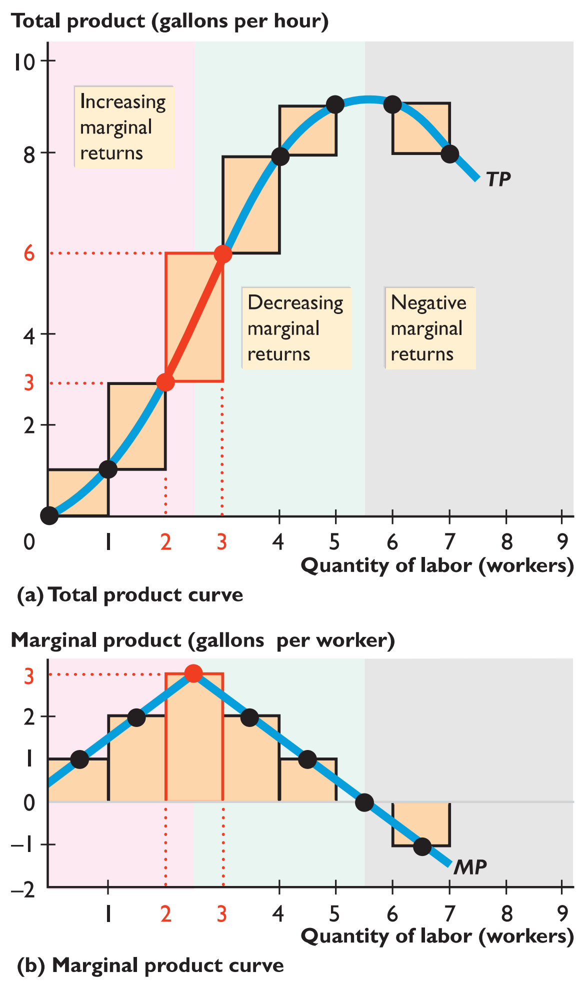
Decreasing marginal returns
When the marginal product of an additional worker is less than the marginal product of the previous worker.

All production processes eventually reach a point of *decreasing* marginal returns. **Decreasing marginal returns** occur when the marginal product of an additional worker is less than the marginal product of the previous worker. Decreasing marginal returns arise from the fact that more and more workers use the same equipment and work space. As more workers are employed, there is less and less that is productive for the additional worker to do. For example, if Samantha hires a

FIGURE 14.3

Total Product and Marginal Product

MyEconLab Animation



Quantity of labor (workers)	0	1	2	3	4	5	6	7
Total product (gallons per hour)	0	1	3	6	8	9	9	8
Marginal product (gallons per worker)		1	2	3	2	1	0	–1

The table calculates marginal product, and the orange bars illustrate it. When labor increases from 2 to 3 workers, total product increases from 3 gallons to 6 gallons of smoothies an hour. So marginal product is the orange bar whose height is 3 gallons (in both parts of the figure).

In part (b), marginal product is graphed midway between the labor inputs to emphasize that it is the result of *changing* inputs. Marginal product increases to a maximum (when 3 workers are employed in this example) and then declines—diminishing marginal product.

fourth worker, output increases but not by as much as it did when she hired the third worker. In this case, three workers exhaust all the possible gains from specialization and the division of labor. By hiring a fourth worker, Sam's produces more smoothies per hour, but the equipment is being operated closer to its limits. Sometimes the fourth worker has nothing to do because the machines are running without the need for further attention.

Hiring yet more workers continues to increase output but by successively smaller amounts until Samantha hires the sixth worker, at which point total product stops rising. Add a seventh worker, and the workplace is so congested that the workers get in each other's way and total product falls.

Decreasing marginal returns are so pervasive that they qualify for the status of a law: the **law of decreasing returns**, which states that

As a firm uses more of a variable factor of production, with a given quantity of fixed factors of production, the marginal product of the variable factor eventually decreases.

Average Product

Average product
Total product divided by the quantity of a factor of production. The average product of labor is total product divided by the quantity of labor employed.

Average product (*AP*) is the total product per worker employed. It is calculated as

$$\text{Average product} = \text{Total product} \div \text{Quantity of labor}.$$

Another name for average product is *productivity*.

Figure 14.4 shows the average product of labor, *AP*, and the relationship between average product and marginal product. Average product increases from 1 to 3 workers (its maximum value) but then decreases as yet more workers are employed. Notice also that average product is largest when average product and marginal product are equal. That is, the marginal product curve cuts the average

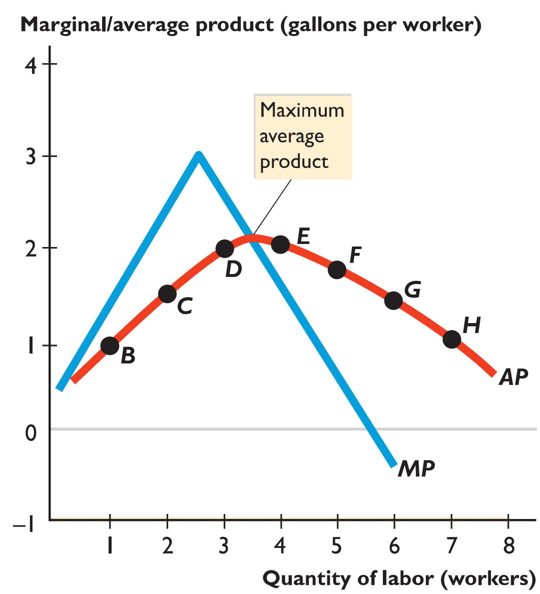
FIGURE 14.4

Average Product and Marginal Product

MyEconLab Animation

The table calculates average product. For example, when the quantity of labor is 3 workers, total product is 6 gallons an hour, so average product is 6 gallons ÷ 3 workers = 2 gallons a worker.

The average product curve is *AP*. When marginal product exceeds average product, average product is increasing. When marginal product is less than average product, average product is decreasing.



Quantity of labor (workers)	0	1	2	3	4	5	6	7
Total product (gallons per hour)	0	1	3	6	8	9	9	8
Marginal product (gallons per worker)	1	2	3	2	1	0	–1	
Average product (gallons per worker)		1.0 *B*	1.5 *C*	2.0 *D*	2.0 *E*	1.8 *F*	1.5 *G*	1.1 *H*

product curve at the point of maximum average product. For employment levels at which marginal product exceeds average product, the average product curve slopes upward and average product increases as more labor is employed. For employment levels at which marginal product is less than average product, the average product curve slopes downward and average product decreases as more labor is employed.

The relationship between average product and marginal product is a general feature of the relationship between the average value and the marginal value of any variable. *Eye on Your Life* looks at a familiar example.

EYE on YOUR LIFE

Your Average and Marginal Grades

Jen, a part-time student, takes one course each semester over five semesters. In the first semester, she takes calculus and her grade is a C (2). This grade is her marginal grade. It is also her average grade—her GPA.

In the next semester, Jen takes French and gets a B (3)—her new marginal grade. When the marginal value exceeds the average value, the average rises. Because Jen's marginal grade exceeds her average grade, the marginal grade pulls her average up. Her GPA rises to 2.5.

In the third semester, Jen takes economics and gets an A (4). Again her marginal grade exceeds her average, so the marginal grade pulls her average up. Jen's GPA is now 3—the average of 2, 3, and 4.

In the fourth semester, she takes history and gets a B (3). Now her marginal grade equals her average. When the marginal value equals the average value, the average doesn't change. So Jen's average remains at 3.

In the fifth semester, Jen takes English and gets a C (2). When the marginal value is below the average value, the average falls. Because Jen's marginal grade, 2, is below her average of 3, the marginal grade pulls the average down. Her GPA falls.

This relationship between Jen's ❶ marginal grade and ❷ average grade is similar to the relationship between marginal product and average product.